

DC/AC Fundamentals

Science

May, 2026

DC/AC Fundamentals (A09891)

Short Title: DC/AC Fundamentals
Faculty Science and Computing
Department Science
Cluster Engineering
Author CWALSH
Credits 5

Level: Introductory

Description of Module / Aims

The student will learn the fundamentals of linear electrical circuits and their components. The student will study the mathematical tools used to represent and analyse electrical circuits and will learn a range of techniques to design DC and AC circuits. There will also be comprehensive study of DC and AC circuits in the laboratory setting, and circuits will also be implemented by the student using National Instruments Circuit Design Suite (Multisim).

Programmes

COMP-0579	BSc (Hons) in Applied Computing (WD_KCOMP_B)	2 / 4 / E
COMP-0579	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	2 / 4 / M
COMP-0579	BSc (Hons) in the Internet of Things (WD_KINTT_B)	2 / 4 / M

Indicative Content

- DC circuit analysis: review of combination DC circuits; loading potential and current dividers; Kirchoff's laws, voltage and current sources; principle of superposition
- Equivalent circuits: Thevenin's theorem; Norton's theorem; maximum power transfer theorem
- AC circuit analysis: introduction to alternating current and voltages; mathematical representations of alternating currents and voltages (sinusoids, rectangular and polar); equivalent circuits
- Phasor mathematics: complex impedance of electrical components; impedance diagrams, analysis of AC circuits; RLC circuits, resonance and Q factor
- Electronic filters: filter classification and applications; passive filters: first order low-pass, high-pass, and band-pass; Bode plots and frequency response

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Analyse a range of DC and AC circuits.
2. Determine the response of resistor, capacitor and inductor networks to sinusoidal waveforms.
3. Apply Thevenin and Norton theorems to simplify a circuit for analysis.
4. Examine a range of first-order filter circuits.
5. Construct, implement and test a range of DC and AC circuits in the integrated practical programme.
6. Model a range of DC and AC circuits using National Instruments Design Suite (Multisim).

Learning and Teaching Methods

- Lectures: Lectures will be used to introduce new topics and their related concepts.
- Practicals: The practical element will allow students to design implement and test electric circuits, interpret and report experimental results.
- Tutorials: Apply problem solving skills to generate solutions to DC and AC circuit problems.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>In-Class Assessment</i>	50%	1,2,3,4
<i>Practical</i>	50%	5,6

Assessment Criteria

<40%: Unable to demonstrate a basic understanding of elementary DC and AC circuits. An understanding of very little of the course content.

40%-49%: Able to provide partial solutions to DC and AC circuit problems.

50%-59%: Able to analyse and provide solutions to circuit problems. Demonstrate an understanding of the mathematical techniques used in circuit analysis.

60%-69%: Able to apply given problem-solving techniques to new similar circuit problems. Demonstrates an ability to design a system from a number of component parts.

70%-100%: All the previous to an excellent level.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	24	
Independent Learning	75	

Supplementary Material

- Floyd, T. _Principles of Electric Circuits: Conventional Current Version_. 9th ed. United States of America: Pearson, 2010.
- Floyd, T. and D. Buchla. _DC/AC Fundamentals A Systems Approach_. United States of America: Pearson, 2013.
- Haskell, R. and D. Hanna. _Solving DC and AC Circuits By Example Using Matlab_. United States of America: LBE Books , 2015.
- Robbins, A. and W. Miller. _Circuit Analysis: Theory and Practice_. 5th ed. United States of America: Delmar Cengage Learning, 2012.

Requested Resources

- SCIENCE LAB: Physics